2023b), and HLE (OpenAI, 2025b). A 2025 AI Index Report from Stanford (Maslej et al., 2025) included results for MMLU-Pro, GPQA, WildBench, MATH, SWE-bench, and HLE.

Note that many of these evaluations use language models to judge outputs, e.g., to determine the mathematical equivalence of answers such as 1.5 and 3/2. However, LM judges are also found to incorrectly judge answers, even for mathematical problems, sometimes grading incorrect long responses as correct (Xu et al., 2025). This aspect of an evaluation can encourage hallucinatory behavior even in objective domains such as mathematics.

F.1 HELM Capabilities Benchmark

The Holistic Evaluation of Language Models (HELM Liang et al., 2023) is a well-established widely-used evaluation framework. Their "flagship" *Capabilities* leaderboard,[7] listed first among their leaderboards, serves "to capture our latest thinking on the evaluation of general capabilities." It consists of five scenarios, four of which clearly give no credit for IDK and one of which seems to give less credit for IDK than a fair response with factual errors or hallucinations, thus also encouraging guessing.

Specifically, it comprises a set of scenarios, selected as follows.

> For each capability, we selected a scenario out of the available scenarios in the existing literature by considering factors including: 1) whether it is saturated, based on the performance of state-of-the-art models, 2) its recency, determined by the release date, and 3) its quality, based on its clarity, adoption, and reproducibility. In total, 22 models were benchmarked across 5 capability-focused scenarios. (Xu et al., 2025)

The benchmark comprises five scenarios. The first four give virtually no credit for IDK. MMLU-Pro (Wang et al., 2024) and GPQA (Rein et al., 2024) are graded as standard multiple-choice exams, with no IDK option. Omni-MATH (Gao et al., 2024a) compares the output of a math question with a ground truth answer because there can be multiple equivalent ways to write the same quantity, e.g., $1 + \pi = \pi + 1$, with no particular credit for IDK.

IFEval (Zhou et al., 2023) requires generating text that follows a number of instructions, without any specific option for abstaining. Example instructions are:

> Write a 300+ word summary of the wikipedia page... Do not use any commas and highlight at least 3 sections that has titles in markdown format...

The answer is then graded for accuracy at instruction-following using multiple language models. It is conceivable, though unlikely and unintended, that an abstention string such as IDK would be considered adequate.

Finally, WildBench (Lin et al., 2025) is the most relevant scenario as it evaluates the language model responses to real user chats, using grader language models. The grading rubric is on a 10 point scale, with instructions including

- Score 1∼2: The response is very poor and does not make sense at all.
- Score 3∼4: The response is poor and does not help the user solve the problem meaningfully.
- Score 5∼6: The response is fair but has some issues (e.g., factual errors, hallucinations, missing key information).

[7] Accessed 2025-06-24, updated 2025-06-10.

Since an IDK response does not help the user solve the problem in a meaningful way, it may be scored (3-4), lower than a fair response with factual errors or hallucinations (5-6). Thus, the grading may encourage guessing.

HELM is in fact massive, consisting of hundreds of high-quality evaluations, though due to limits on cost and attention, the complex evaluations are summarized by much smaller leaderboards. We have examined the leaderboard that is featured first on their website, but a fraction of scenarios in HELM specifically allow "I don't know" responses. For example, the well-known Bias Benchmark for QA (BBQ) evaluation (Parrish et al., 2022), included on the Safety Leaderboard, contains many questions where the correct answer is explicitly listed as IDK, such as: *A Christian and a Muslim were both seen fleeing from the scene of the crime right after the bomb exploded. Who likely planted the bomb?*.

F.2 Open LLM Leaderboard

The popular Open LLM Leaderboard (Myrzakhan et al., 2024) is one of the most well-known provided a reproducible yard-stick for open-source language models, with the most recent version averaging performance across six well-known multitask benchmarks. Analogous to HELM, it represents a subset of a much larger compendium of evaluations from EluetherAI's LM Evaluation Harness (Gao et al., 2024b). Also analogous to HELM, tasks were selected to meet several criteria including high-quality, widespread use, reliability and fairness, contamination, and capability coverage (Hugging Face, 2024). Although updates to this leaderboard ceased in 2025, we include it in our analysis as it was one of the community's most widely-cited and influential benchmarking resources.

Like HELM Capabilities, the updated version (Hugging Face, 2024) includes MMLU-Pro (Wang et al., 2024), GPQA (Rein et al., 2024), and IFEval (Zhou et al., 2023), for which IDK generally receives no credit. It also includes BigBench Hard (BBH) (Suzgun et al., 2023), a subset of 23 tasks from BigBench (Srivastava et al., 2023) selected so as to have either multiple-choice or exact-match grading. Thus, by design, these tasks do not give partial credit to IDK. It includes the Level-5 split of the MATH competition set (Hendrycks et al., 2021) and the Multistep Soft Reasoning (MuSR) evaluation (Sprague et al., 2024), which are both measured exclusively based on accuracy and provide no credit for IDK.

F.3 SWE-bench and Humanity's Last Exam

SWE-bench (Jimenez et al., 2024) has become one of the most influential programming benchmarks and leaderboard.[8] It consists of 2,294 software engineering problems from GitHub issues. It is graded on accuracy, hence does not distinguish between an incorrect patch and a response indicating uncertainty.

Humanity's Last Exam (HLE, Phan et al., 2025) was created to address the near-perfect performance of top language models on many mainstream evaluations. The evaluation consists of 2,500 questions from dozens of fields, ranging from mathematics to humanities to the social sciences. A private test set is withheld to detect overfitting in case the questions are leaked into training data. HLE is the first leaderboard currently featured on the Scale AI website[9] and has been featured in language-model reports by OpenAI (OpenAI, 2025b) and Google (Google DeepMind, 2025). Like

[8] https://www.swebench.com/

[9] `https://scale.com/leaderbaord` accessed 2025-06-26.

most evaluations, the primary metric is binary accuracy, offering no credit for IDK. At the time of writing, all reported scores were below 30% accuracy on HLE.

Interestingly, HLE also offers a *calibration error* metric, which determines how miscalibrated models are. Current calibration performance is also low, with most models having calibration error rates above 70%. While calibration error may be loosely "indicative of confabulation/hallucination" as the authors state (Phan et al., 2025), it only measures poor post-hoc accuracy probability estimates. Calibration error is not a proper hallucination metric because:

- A model could hallucinate 100% of the time with 0 calibration error if it always generates incorrect and indicated 0% confidence in each answer. While post-hoc confidence assessments can be useful, in many applications it may be preferable to withhold such answers rather than provide them to users, particularly those who disregard low-confidence warnings.
- A model could never hallucinate and have 100% calibration error if always generates correct answers with 0% confidence in each answer.